DATA STRUCTURE AND PROGRAM DESIGN LAB – 06

6. Implement Binary search tree(BST) with following Menu operations.

1. Search an element in BST(Display NULL if not found, If found Display Found)
2. Insert an element in BST
3. Delete leaf element in BST
4. Exit

SAMPLE OUTPUT:

```
--- BST MENU ---
1. Search an Element
2. Insert an Element
3. Delete Leaf Element
4. Exit
Enter your choice: 2
Enter value to insert: 50
Inserted.

--- BST MENU ---
1. Search an Element
2. Insert an Element
3. Delete Leaf Element
4. Exit
Enter your choice: 2
Enter value to insert: 30
```

```
Enter your choice: 2
Enter value to insert: 20
Inserted.

--- BST MENU ---
1. Search an Element
2. Insert an Element
3. Delete Leaf Element
4. Exit
Enter your choice: 2
Enter value to insert: 40
Inserted.

--- BST MENU ---
1. Search an Element
2. Insert an Element
```

```
3. Delete Leaf Element
4. Exit
Enter your choice: 2
Enter value to insert: 60
Inserted.

--- BST MENU ---
1. Search an Element
2. Insert an Element
3. Delete Leaf Element
4. Exit
Enter your choice: 2
Enter value to insert: 80
Inserted.

--- BST MENU ---
```

```
1. Search an Element
2. Insert an Element
3. Delete Leaf Element
4. Exit
Enter your choice: 1
Enter value to search: 40
Found

--- BST MENU ---
1. Search an Element
2. Insert an Element
3. Delete Leaf Element
4. Exit
Enter your choice: 1
Enter value to search: 100
NULL (Element not found)
```

```
--- BST MENU ---
1. Search an Element
2. Insert an Element
3. Delete Leaf Element
4. Exit
Enter your choice: 3
Enter leaf value to delete: 20
Leaf node 20 deleted.

--- BST MENU ---
1. Search an Element
2. Insert an Element
3. Delete Leaf Element
4. Exit
Enter your choice: 3
Enter leaf value to delete: 30
```

Speakers (Realtek(R) Audio): 100%

```
Cannot delete 30 (Not a leaf node)

--- BST MENU ---
1. Search an Element
2. Insert an Element
3. Delete Leaf Element
4. Exit
Enter your choice: 4
```